

Rule 1: Cross-merchandising

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Spaghetti → Pasta Sauce
3. Question: Customers buying spaghetti frequently buy pasta sauce. How should we physically present these products in the store?
4. Retail policy/rule: Cross-merchandising means displaying complementary products together or through secondary displays so that the shopping mission becomes easier and additional purchases may increase.

Rule 2: Store-layout management

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Chips → Salsa
3. Question: Because chips and salsa have a strong association, should they necessarily be placed next to each other?
4. Retail policy/rule: Strong complements may be kept near for convenience, but they may also be separated deliberately to increase store traversal and expose customers to more merchandise.

Rule 3: Cross-selling

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Shrimp → Frozen Vegetables
3. Question A customer has already put shrimp into the basket. What should the retailer recommend next?
4. Retail policy/rule: Cross-selling uses the antecedent purchase as a trigger for suggesting a complementary product.

Rule 4: Product bundling

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Spaghetti → Pasta Sauce
3. Question Should we sell spaghetti and pasta sauce as one combined offer?
4. Retail policy/rule: Strongly associated items can be offered as bundles or meal solutions (for example, at McDonalds), provided the economics justify the bundle.

Rule 5: Promotional planning

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Burgers → French Fries and reverse French fries to Burger
3. Question: If we want to promote burgers, should we also discount French fries?

Rule 6: Coupon targeting

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Shrimp → Frozen Vegetables
3. Question If a customer buys shrimp today, what coupon could we give for the next visit?
4. Retail policy/rule: Transaction history can determine which complementary-item coupon is most relevant to a customer.

Rule 7: Recommendation systems

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Shrimp → Frozen Vegetables or Spaghetti → Pasta Sauce
3. Question If this were an online grocery website, what should appear under “You may also like”?
4. Retail policy/rule: Association rules can provide simple next-best-product recommendations.

Rule 8: Inventory coordination

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Cereal → Milk
3. Question Suppose cereal goes on a major promotion next week. Should the inventory manager change only the cereal order?
4. Retail policy/rule: Promotion or demand for one item may create additional demand for associated products. Replenishment should therefore consider related SKUs.

Rule 9: Assortment planning

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Compare several strong rules
3. Question: Suppose shelf space is limited. Which products should the retailer definitely continue carrying?

Spaghetti and Pasta Sauce — Support 30%, Confidence 75%, Lift ≈ 2.14 . This is the **highest-volume strong association** in the data.

Cereal and Milk — Support 20%, Confidence 80%, Lift = 2.00. Very strong and commercially intuitive.

Shrimp and Frozen Vegetables — Support 15%, Confidence 75%, Lift = 3.00. Lower volume, but an exceptionally strong affinity.

Burgers and French Fries — Support 18%, Confidence 60%, Lift = 2.00.

Chips and Salsa — Support 15%, Confidence 60%, Lift = 3.00.

4. Retail policy/rule: Products should not be evaluated only by individual sales. Some products contribute significantly to valuable baskets through their associations.

Rule 10: Category management

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: chicken → Spaghetti
3. Question Chicken belongs to meat and spaghetti belongs to packaged food. Why should managers of these two categories care about each other?
4. Retail policy/rule: Customers shop according to needs and consumption missions rather than the retailer's departmental structure. Categories may need to be managed jointly.

Rule 11: Promotion-effectiveness policy

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Burgers → French Fries, before and after a hypothetical promotion
3. Question If we run a burger promotion, how will we know whether it changed the customer's basket rather than merely increasing burger sales?
4. Retail policy/rule: Compare association strength, basket size, margin and attached-product sales before and after the promotion.

Rule 12: Personalized marketing

1. Set support and confidence: Support 10%, Confidence 40%
2. Rule to observe: Compare Eggs → Mineral Water with Mineral Water → Eggs
3. Question Which rule would you use if the customer has just purchased eggs?
4. Retail policy/rule: Recommendation direction matters. Use the conditional relationship relevant to the customer's observed behavior. When Egg is purchased offer mineral water but not the reverse.

Rule 13: Shelf-space allocation

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Chips → Salsa or Spaghetti → Pasta Sauce
3. Question If these products have strong affinities, should they receive additional shelf visibility?
4. Retail policy/rule: Association information can complement sales volume when allocating facings, end caps and secondary display space.

Forms of personalized Marketing

- **Real-time recommendation:** “You may also like mineral water.”
- **Personalized coupon:** Buy mineral water at 10% off after purchasing eggs.
- **Next-visit targeting:** Send a mineral-water offer to recent egg purchasers.
- **App/homepage personalization:** Show mineral water higher in the recommendation list for customers who frequently buy eggs.
- **Loyalty-program targeting:** Create different offers for different basket histories.

Rule 14: Private-label strategy

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Spaghetti → Pasta Sauce
3. Question: Suppose spaghetti is a leading national brand. Could the retailer introduce its own private-label pasta sauce beside it?
4. Retail policy/rule: Strong complementary demand can reveal opportunities for retailer-owned products around popular traffic-generating items.

Rule 15: Loss-leader strategy

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Burgers → French Fries
3. Question: Suppose burgers have a very low margin, but French fries have a high margin. Could the retailer deliberately discount burgers?
4. Retail policy/rule: A low-margin or discounted product can be worthwhile if it reliably stimulates profitable associated purchases.

Rule 16: Basket-profitability management

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Burgers → French Fries
3. Question: Which is more important: burger profitability or the profitability of the complete basket?
4. Retail policy/rule: Evaluate the total contribution margin generated by the shopping basket, not merely individual SKU margins.

Rule 17: Stock-out management

1. Set support and confidence: Support 10%, Confidence 50%
2. Rule to observe: Cereal → Milk and the reverse
3. Question What happens to cereal sales if milk goes out of stock? Or to milk sales if cereal is unavailable?
4. Retail policy/rule: A stock-out can have spillover effects on associated products, so service levels for strongly connected products may need coordinated management.

Rule 18: Seasonal merchandising

1. Set support and confidence: Support 10%, Confidence 40–50%
2. Rule to observe: Compare rules across hypothetical summer/winter or festival/non-festival periods
3. Question Would you expect the same basket relationships during summer, winter, weekends and festivals?
4. Retail policy/rule: Association patterns can be calculated separately by period so assortments, displays, promotions and inventory reflect changing shopping missions.